CAUSTICS & WATER SURFACE SIMULATION

ADVANCED GRAPHICS - TRACK 4

STUDENT INFO:

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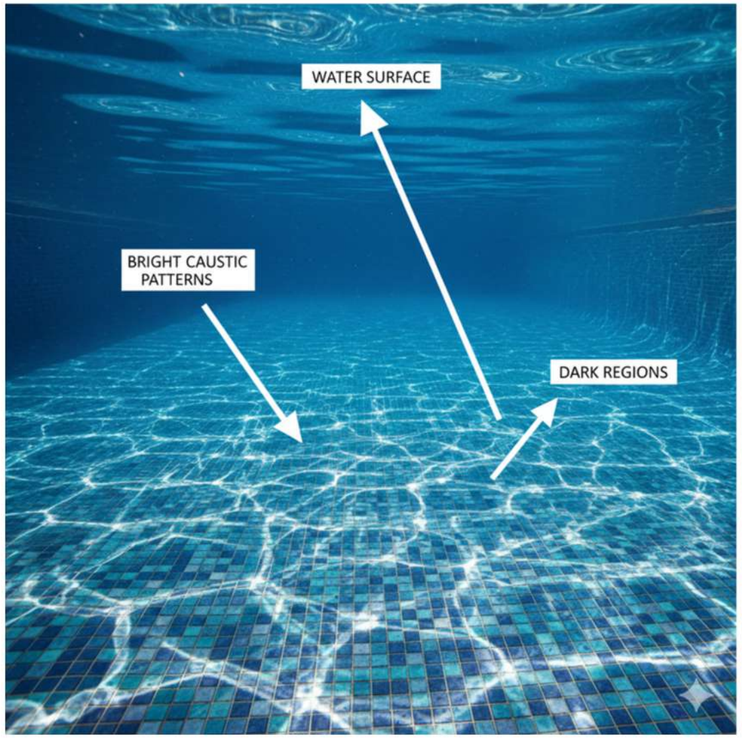
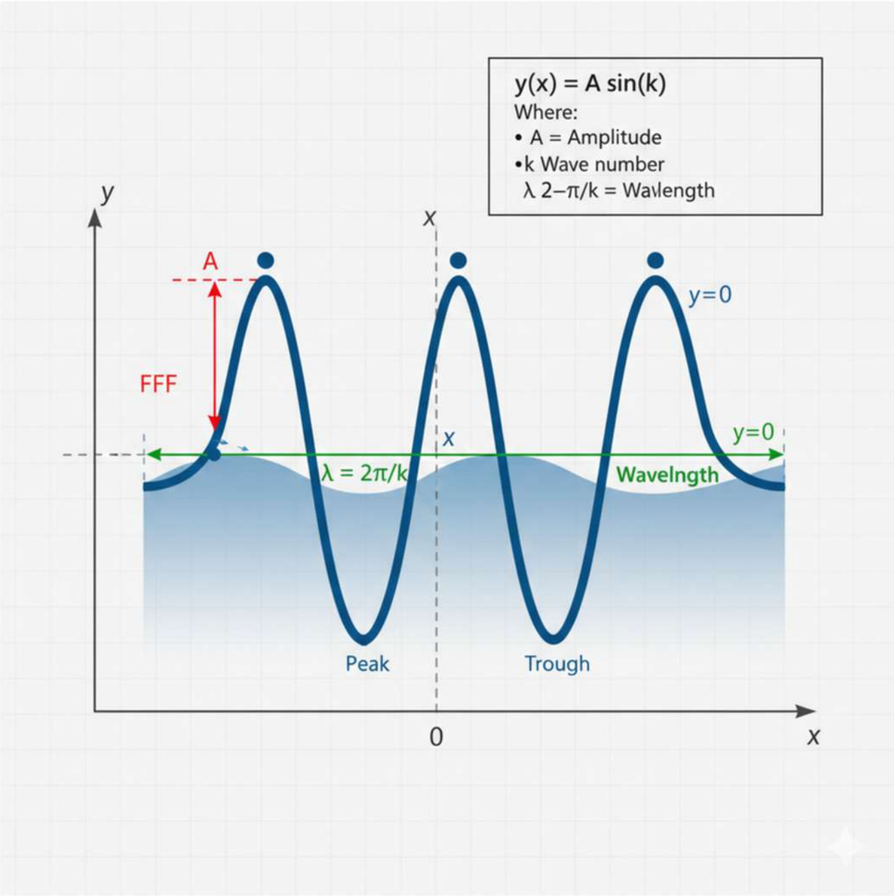
PHYSICS: LIGHT REFRACTION

Light coming from above the pool bends at every little ripple of the water. The bending depends on the slope of the surface. Rays that bend toward each other create bright patterns on the pool floor; rays that spread out create darker areas. The whole process is determined by the surface geometry and the laws of refraction



WATER SURFACE MODEL



Sinusoidal Wave Function:

$$y(x) = A \sin(kx)$$

Parameters:

- A (Amplitude): Wave height
- k(Wavenumber):Wave frequency/spacing
- x: Horizontal position

MATHEMATICS: RAYS AND NORMALS

INCIDENT RAY (VERTICAL LIGHT):

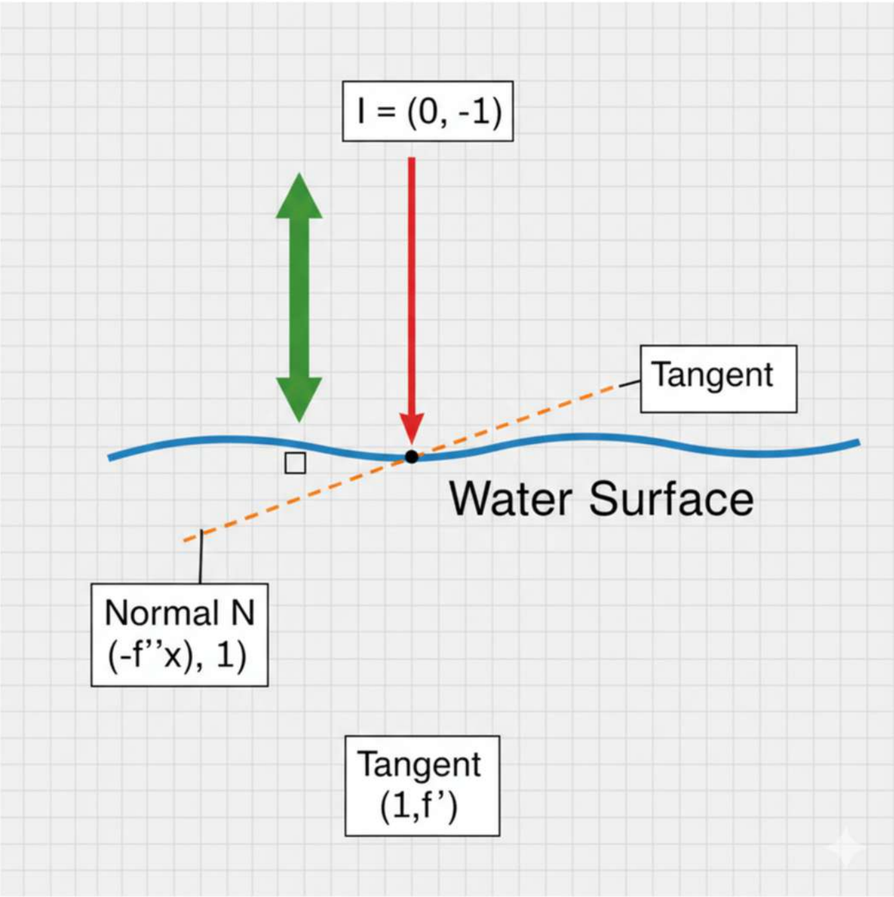
I = (0, -1)

→ Light comes straight down (simplifies calculations)

SURFACE NORMAL VECTOR:

Surface slope: **f'(x) = dy/dx**

Normal vector: **N = (-f'(x), 1)**



why (-f'(x), 1)?

Surface equation

y = f(x)

Tangent vector (surface slope)

Tangent = (1, f'(x))

Normal is Perpendicular to Tangent

<u>Perpendicularity condition</u>

a·c + b·d = 0

Given

Tangent = (1, f'(x))

Normal = (a, b)

Solve

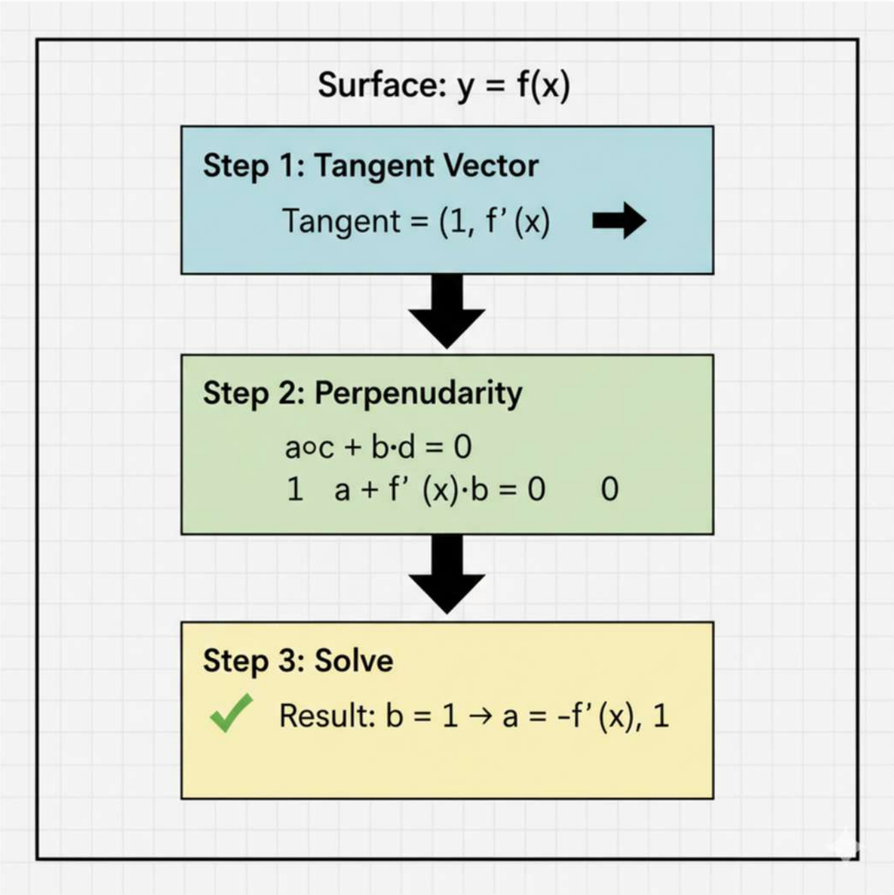
a + f'(x)·b = 0·1

a = -f'(x)·b → Choose b = 1 (convention) Then a = -f'(x) →

Result

Normal = (-f'(x), 1)

(Then normalize to unit length)

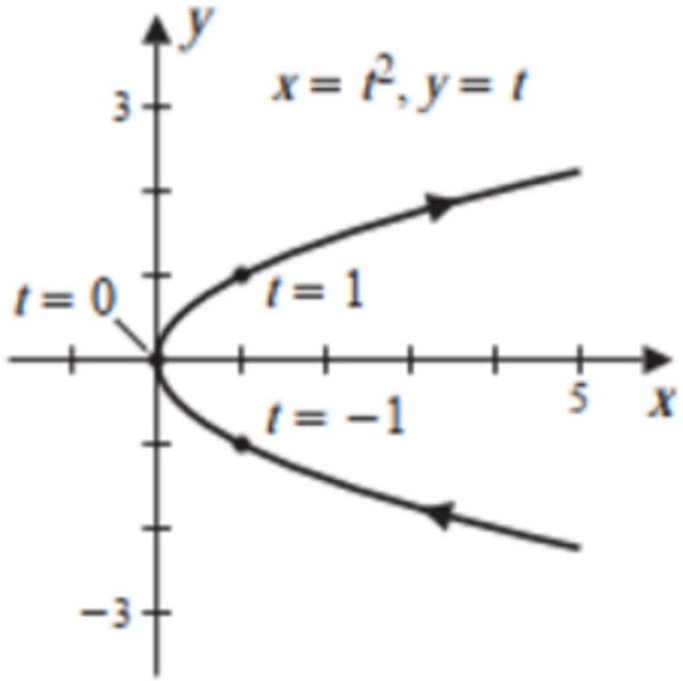


MATHEMATICS: RAY PROPAGATION

Ray Equation:

$P(t) = P_0 + t \cdot T$

- P_0: Entry point on water surface
- T: Refracted ray direction
- t: Distance parameter (how far along the ray)



FINDING BOTTOM INTERSECTION:

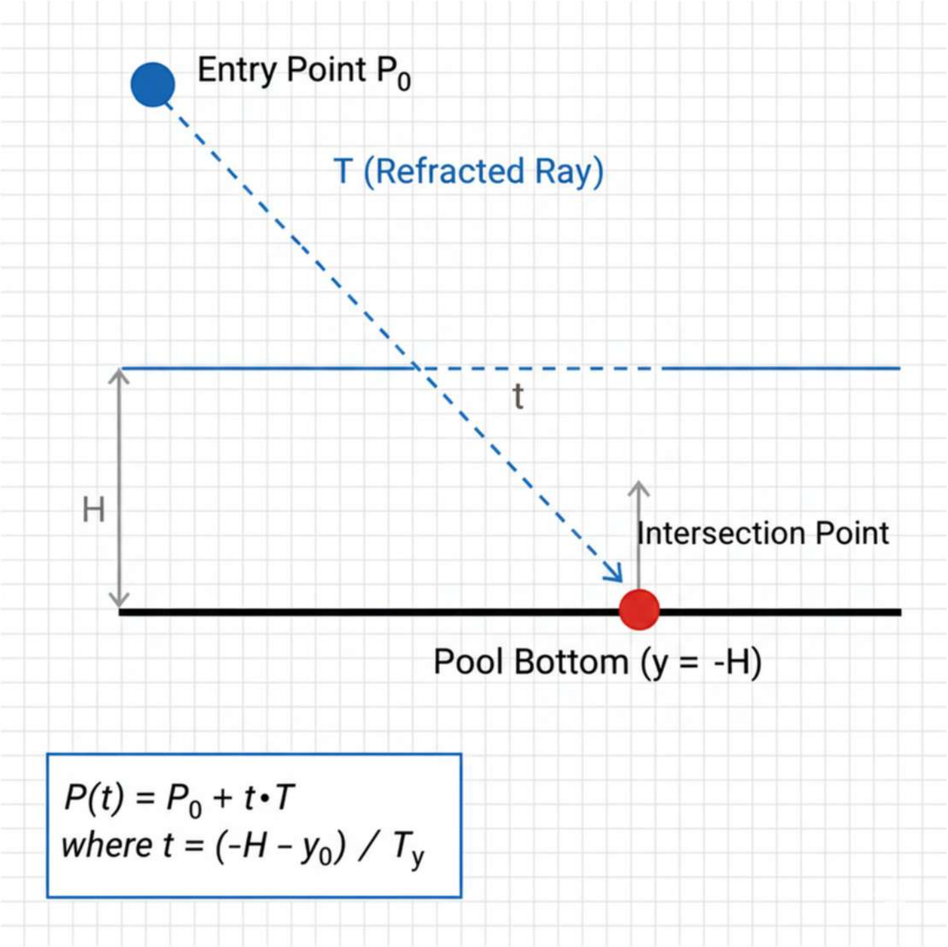
Pool bottom: $y = -H$

Solve: $y_0 + t \cdot T_y = -H$

Calculate: $t = (-H - y_0) / T_y$

<u>Caustics Formation:</u>

- Ray convergence → Bright regions (high intensity)
- Ray divergence → Dark regions (low intensity)



THE FULL PICTURE

Step-by-Step Process:

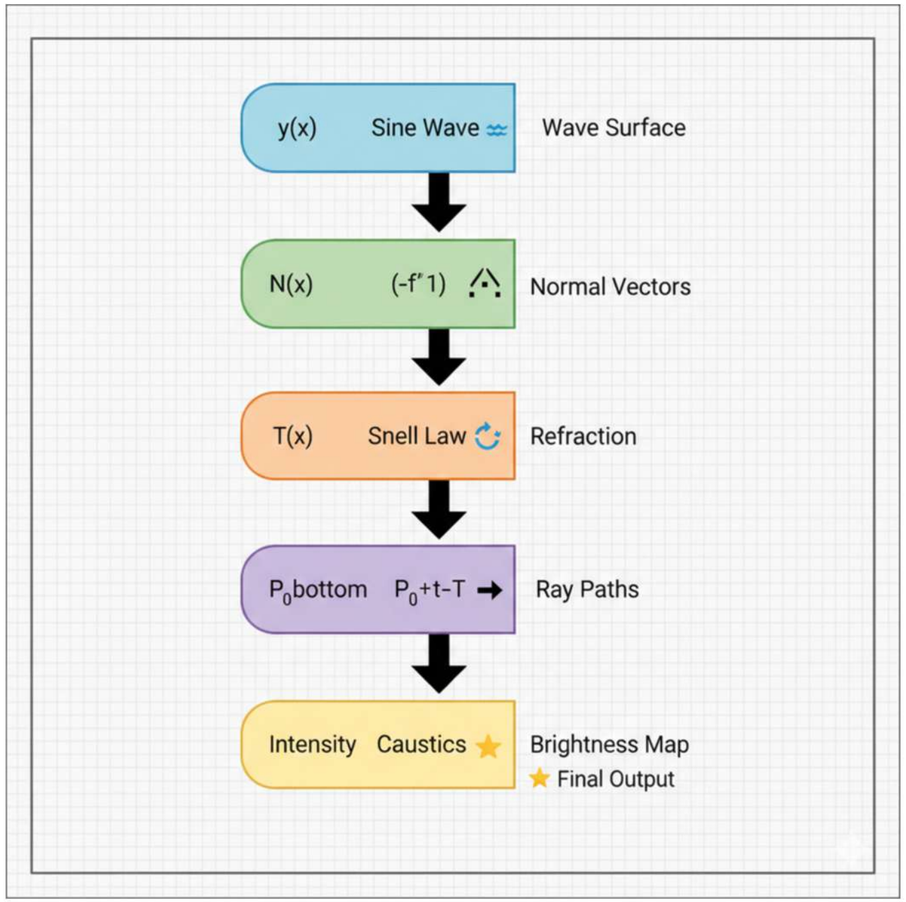
1. Waves define Surface Normals
2. Normals define Refraction Angles
3. Refraction defines Ray Paths
4. Ray Density defines Brightness

Mathematical Chain:

y(x) → N(x) → T(x) → P_bottom(x) → Intensity(x)

Energy Conservation:

- Light energy is conserved
- Where rays converge = higher intensity
- Where rays spread = lower intensity



PROJECT OBJECTIVES

Main Goal: Create a computer simulation that generates realistic caustic patterns

Input:

- Procedural water surface (Sine waves or Perlin Noise)
- Light ray parameters
- Pool depth

Output:

- Image showing the "web of light" on the pool floor
- Visual representation of ray convergence

Deliverable:

- Working implementation
- Generated caustics image

IMPLEMENTATION STRATEGY

Algorithm Steps:

Step 1: Generate Water Surface

- Use sine waves: $y = A \sin(kx)$
- Or Perlin noise for complexity

Step 2: Ray Tracing

- For each surface point:
  - Calculate normal vector
  - Apply Snell's Law (vector form)
  - Trace refracted ray to bottom

Step 3: Additive Blending

- Initialize brightness map (all pixels = 0)
- For each ray intersection:
  - brightness[x, y] += intensity
- Normalize final image

EXPECTED OUTPUT

Visual Deliverable:

- High-quality image showing caustic patterns
- "Web of light" at pool bottom
- Realistic intensity variations

Comparison:

- Before: Flat surface → uniform lighting
- After: Wavy surface → caustic patterns

Success Criteria:

- Visible bright caustic lines
- Physically plausible pattern distribution
- Smooth intensity gradients

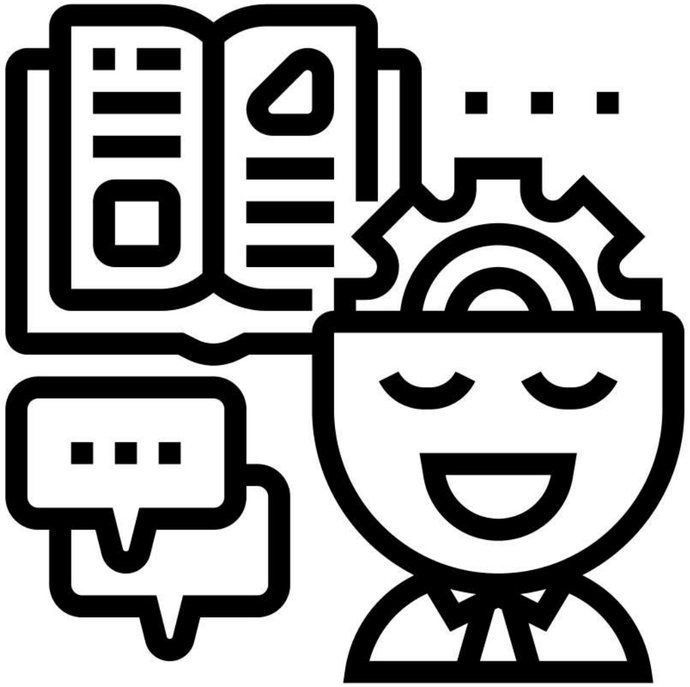
CONCLUSION

Summary:

- Caustics are light patterns created by refraction
- Mathematical model: Snell's Law + Ray Tracing
- Implementation: Surface generation + Ray intersection

Applications:

- Computer Graphics & CGI
- Video Games (realistic water rendering)
- Visual Effects in films
- Architectural visualization



THANK YOU FOR YOUR ATTENTION